Covariate-adjusted hot and cold spots of urban pests with eigenvector spatial filtering

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Abstract

Municipalities that control urban pests need to decide where to act first, and citizen reports of sightings are often the only data available. Reports, however, mostly follow where people live. We define a hot spot as an area with more reports than its local characteristics predict, shared with its neighbours, and propose a procedure to map such hot and cold spots and to rank areas for intervention. The procedure tests the residuals of a regression on the covariates for spatial structure and stops if there is none; estimates the spatial component by eigenvector spatial filtering, with the smooth eigenvectors of the contiguity matrix and a ridge penalty chosen by cross-validation; labels each area hot, cold or neutral with a permutation band; and orders the hot areas into a priority list. In a simulation study with nine scenarios, the procedure was as precise as the Getis–Ord G_i^* statistic and the local Moran's I with strong clusters, found clusters of different sizes more often, and was less precise with weak compact clusters; the test before mapping reduced the share of the map falsely labelled hot from about 5% to at most 2% for all methods. Applied to cockroach sightings in Madrid in 2010–2013, with nested models for exposure, environmental risk factors and rat reports, the procedure found hot zones that persisted across summers and differed from those of rats once exposure was accounted for. Evaluated on the following summer, ranking by counts captured the most reports, and ranking the residuals of single cells captured more of the excess over the covariates than

the smoothed map; we suggest using the hot zones to decide where to act and the residuals to order cells within them.

1 Introduction

Local governments control urban pests with small teams, limited budgets and a restricted set of resources such as traps, chemicals, staff and vehicles, so they must decide where to act first. The decision matters for public health. Cockroaches carry pathogens and parasites (Kalantari *et al.*, 2023; Abdullah *et al.*, 2024) and are associated with allergic sensitisation and asthma (Liccardi *et al.*, 2023), and the cost of pest control is substantial: 61 of the 142 National Health Service trusts in the United Kingdom spent £3.7 million on it between 2021 and 2023 (Liberal Democrats, 2024). A municipality that receives citizen reports of sightings, such as the vector control unit of Madrid, can use them to decide where to place traps or to inspect first. This paper asks how to build such a priority list from the reports and from what is known about each area of the city.

The usual tools for this question are hot-spot statistics. The Getis–Ord statistic G_i^* (Getis and Ord, 1992; Ord and Getis, 1995) compares the sum of a variable around each area with its expected value, and the local Moran's I (Anselin, 1995) detects areas whose values resemble those of their neighbours. Scan statistics (Kulldorff, 1997) search for the circular window with the largest excess of cases, and disease-mapping models (Besag *et al.*, 1991) smooth relative risks with a spatially structured random effect. For urban pests in Madrid, Tamayo-Uria *et al.* (2014) and Ayyad *et al.* (2018, 2019) modelled the intensity of rat sightings as a function of environmental risk factors and produced risk maps from the fitted values.

Three features of report data are not fully addressed by these tools. First, the number of reports in an area depends on how many people live there and can see a pest, so a map of counts mostly shows where people live; a useful hot spot is an area with more reports than its characteristics predict. Second, local statistics are computed for every area of the map, so some areas are labelled hot by chance even when there is no spatial structure; the error control should apply to the map as a whole. Third, a priority list should be judged by what it achieves, with the true hot areas known in simulation and with future reports in the application, and it should be compared with the simpler alternatives that a practitioner would otherwise use.

We propose a procedure that addresses these three points. A hot spot is defined as an area where the variation not explained by the covariates is positive and shared with its neighbours (Section 2). The procedure first tests whether the residuals of the regression on the covariates

have spatial structure, with Moran's I and its exact moments (Tiefelsdorf and Griffith, 2007), and stops if they do not. It then estimates the spatial component with eigenvector spatial filtering (ESF) (Griffith, 2000, 2003), using the smooth eigenvectors of the contiguity matrix and a ridge penalty chosen by cross-validation, labels each area hot, cold or neutral with a permutation band, and orders the areas into a priority list. Each of these choices is motivated with simulations in Section 3.2.

The contributions of the paper are the following. We give a definition of covariate-adjusted hot and cold spots and a procedure to estimate them with a test before mapping. We compare the procedure with G_i^*, local Moran's I and rankings without spatial smoothing in a simulation study with nine scenarios that vary the size, shape and number of the clusters and include a cold area, a missing covariate and data without clusters. We apply the procedure to cockroach sightings in Madrid in 2010–2013, with nested models that separate exposure (how many people can report), environmental risk factors and reporting habits (rat reports in the same areas), and we evaluate the priority lists on the following summer's reports. The method is implemented in an R package available in the project repository.

The rest of the paper is organised as follows. Section 2 describes the procedure and the rationale for each step. Section 3 presents the simulation study and Section 4 the application to Madrid. Section 5 discusses when to use the procedure, its limitations and possible extensions.

2 Method

A hot spot is an area with more events than its local characteristics predict, and with neighbours that share this excess. This definition depends on the covariates: the same data can show different hot spots under different sets of covariates, and Section 4 uses this to separate where people report from where the environment predicts more pests. Formally, let $s_1, \ldots, s_n$ be a partition of region R. For each cell i we observe: the number of events w_i and a vector of k covariates z_i. Spatial relations between cells are described by an adjacency matrix $A = (a_{ij})$, with $a_{ij} = 1$ when cells i and j share a border or a corner (queen contiguity) and $a_{ii} = 0$, for all $i, j \in \{1, \ldots, n\}$. Let y_i be the standardised value of $\log(1 + w_i)$, the matrix of standardised covariates $Z \in \mathbb{R}^{n \times k}$ and $X = (\mathbf{1}, Z)$. We write

$$y = X\beta + f + \varepsilon, \tag{1}$$

where $\beta \in \mathbb{R}^{k+1}$ are the covariate effects, $f = (f_1, \ldots, f_n)^\top \in \mathbb{R}^n$ is a spatially structured component and $\varepsilon \in \mathbb{R}^n$ is unstructured noise. The component f collects the variation in events

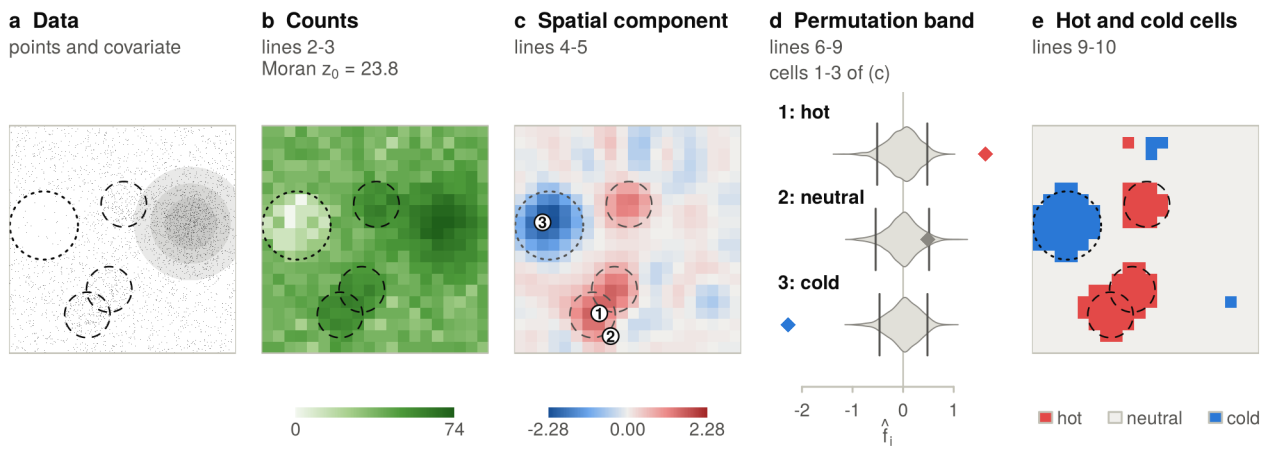


Figure 1: The procedure on one simulated dataset (scenario K of Section 3, strong version, 20×20 grid): three clusters of 200 points (dashed circles), a cold area that keeps 20% of its points (dotted circle), 2000 uniform points and 3000 points that follow a covariate (grey area). (a) The points. (b) Counts per cell, in green; the covariate area has the largest counts but is explained by the covariate. The test of Step 1 rejects (z_0 is the standardised Moran's I). (c) The estimated spatial component $\hat{f}$ (red positive, blue negative): the covariate area has disappeared and the clusters and the cold area remain. (d) The permutation band for the three cells marked in (c): distribution of $\hat{f}_i^{(b)}$ over $B = 999$ permutations (grey), its 5% and 95% quantiles (vertical ticks), and the observed $\hat{f}_i$ (diamond). Cell 2 has a positive $\hat{f}_i$ inside the band and is labelled neutral. (e) The resulting hot and cold cells. Line numbers refer to Algorithm 1.

that the covariates do not explain and that is shared by neighbouring cells. We define a *hot spot* as a cell with $f_i > 0$ and a *cold spot* as a cell with $f_i < 0$, in both cases, when f_i is larger in absolute value than what spatially unstructured noise would produce.

In order to determine the hot spots of region R we consider a procedure with four steps: a test for residual spatial structure, an estimate of f by eigenvector spatial filtering, a permutation band that labels each cell as hot, cold or neutral, and the construction of zones and of a priority list. Algorithm 1 summarises it. Figure 1 illustrates the procedure on a simulated dataset.

Step 1: Residual spatial structure

Consider the regression model without the spatial component, that is, model (1) under the null hypothesis $H_0 \colon f = 0$,

$$y = X\beta + \varepsilon, \tag{2}$$

where $M_X = I - X(X^\top X)^{-1}X^\top$, and $u = y - X\hat{\beta}_{\text{OLS}} = M_X y$ are the residuals of the

ordinary least squares (OLS) regression of y on X, with $\hat{\beta}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top y$. Then, the Moran's I of the residuals,

$$I = \frac{u^\top A u}{u^\top u}, \tag{3}$$

is compared with its exact mean and variance under the null hypothesis of no spatial correlation, which for regression residuals depend on X and A only (Tiefelsdorf and Griffith, 2007). With $T_1 = \mathrm{tr}(M_X A M_X)$, $T_2 = \mathrm{tr}(M_X A M_X M_X A M_X)$ and $d = n - \mathrm{rank}(X)$,

$$\mathrm{E}(I) = \frac{T_1}{d}, \qquad \mathrm{Var}(I) = \frac{2\,(d\,T_2 - T_1^2)}{d^2(d+2)}, \qquad z_0 = \frac{I - \mathrm{E}(I)}{\sqrt{\mathrm{Var}(I)}}. \tag{4}$$

If the one-sided p-value of z_0 is larger than α (we use $\alpha = 0.05$), the procedure stops and every cell is labelled neutral: the covariates explain the spatial pattern of the sightings and there is no evidence of hot or cold spots. Otherwise the procedure continues.

Step 2: Spatial component

Let $E = (e_1, \ldots, e_n)$ orthonormal eigenvectors and $\Delta = \mathrm{diag}(\delta_1 \geq \cdots \geq \delta_n)$ eigenvalues of A so that $A = E \Delta E^\top$. Hot and cold spots are positively autocorrelated by definition, so we use only the candidate set

$$\mathcal{C} = \{\, j : \ \delta_j \geq c\,\delta_1 \,\}, \qquad \text{with } c = 0.25, \tag{5}$$

which is the threshold suggested by Griffith (2003). On a regular queen grid this keeps about one fifth of the eigenvectors. With $E_{\mathcal{C}}$ the matrix of candidate eigenvectors, model (1) becomes

$$y = X\beta + E_{\mathcal{C}}\gamma + \varepsilon, \tag{6}$$

and we estimate it by ridge regression (Hoerl and Kennard, 1970), penalising the eigenvector coefficients but not the covariate effects:

$$(\hat{\beta}, \hat{\gamma}) = \arg\min_{\beta,\gamma} \ \frac{1}{2n} \left\| y - X\beta - E_{\mathcal{C}}\gamma \right\|_2^2 + \frac{\lambda}{2} \left\| \gamma \right\|_2^2, \qquad \hat{\gamma} = \left(E_{\mathcal{C}}^\top M_X E_{\mathcal{C}} + n\lambda I \right)^{-1} E_{\mathcal{C}}^\top M_X\, y. \tag{7}$$

We use ridge instead of the Moran's I Lasso of Barde *et al.* (2025) because our aim is to map f, not to select eigenvectors: with the Lasso penalty $|z_0|^{-2}$, weak clusters often lead to an empty selection and $\hat{f} = 0$. In the simulation study, ridge gave the same precision as the Lasso with strong clusters and a higher one with weak clusters (Section 3.2).

The penalty λ is chosen by ten-fold cross-validation (the value with the smallest cross-validated error), and the model is fitted with the `glmnet` package (Friedman *et al.*, 2010).

The estimated spatial component is

$$\hat{f} = E_{\mathcal{C}}\hat{\gamma} - \overline{E_{\mathcal{C}}\hat{\gamma}}\,\mathbf{1}, \tag{8}$$

centred because the leading eigenvector of a connected graph is nearly constant and therefore almost collinear with the intercept. When the covariates are close to orthogonal to the candidate eigenvectors, $E_{\mathcal{C}}^{\top} M_X E_{\mathcal{C}} \approx I$ and ridge shrinks all coefficients by the same factor, so that $\hat{f}$ is, up to a scale, the projection of the residuals u on the smooth patterns in $\mathcal{C}$.

Step 3: Hot, cold and neutral cells

Let $b \in \{1, \ldots, B\}$ and Π_b a random permutation matrix. Following Freedman and Lane (1983), we permute the OLS residuals $y^{(b)} = X\hat{\beta}_{\text{OLS}} + \Pi_b u$, and refit (7) with the same candidate set and the same λ, which gives $\hat{f}^{(b)}$. With $\mathcal{C}$ and λ fixed, the ridge estimate is linear in y: by (7) and (8),

$$\hat{f} = Sy, \qquad S = \left(I - \tfrac{1}{n}\mathbf{1}\mathbf{1}^{\top}\right) E_{\mathcal{C}} \left(E_{\mathcal{C}}^{\top} M_X E_{\mathcal{C}} + n\lambda I\right)^{-1} E_{\mathcal{C}}^{\top} M_X, \tag{9}$$

and $SX = 0$ because $M_X X = 0$. (The `glmnet` package standardises the columns before penalising, which changes the scale of λ; since the candidate eigenvectors have unit norm and mean close to zero, S keeps the form (9) up to this rescaling, and our implementation uses the exact `glmnet` version.) Hence $\hat{f}^{(b)} = S\,\Pi_b u$: the refit does not need to be carried out, and the whole band is obtained from one $n \times n$ matrix S applied to the B permuted residual vectors.

For each cell i, let q_i^{lo} and q_i^{hi} be the 5% and 95% quantiles of $\hat{f}_i^{(1)}, \ldots, \hat{f}_i^{(B)}$. Cell i is labelled

$$\text{hot if } \hat{f}_i > q_i^{\text{hi}}, \qquad \text{cold if } \hat{f}_i < q_i^{\text{lo}}, \qquad \text{neutral otherwise.} \tag{10}$$

The band is pointwise: when there is no spatial structure, each cell has a 5% chance of being labelled hot, so a map of n cells would contain about $0.05\,n$ false hot cells. The test of Step 1 protects the map as a whole against this: the band is only computed when the test rejects, so when there is no spatial structure anywhere, the probability that any cell is labelled hot is at most α. When the test rejects because of a true cluster, cells outside it keep the pointwise 5% rate.

In the simulation study and in the application, we use $B = 999$, the usual choice for permutation tests, for which the 5% and 95% quantiles fall on order statistics and the

Monte Carlo standard error of a tail probability of 0.05 is about 0.007. With smaller values the band is slightly too narrow: with the usual (linear interpolation) sample quantile, the probability that a cell is labelled hot under the null hypothesis is about 5.1% for $B = 999$, but 5.5% for $B = 200$ and 5.9% for $B = 100$.

Step 4: Zones and priority list

Hot (cold) zones are the connected components of the hot (cold) cells in the graph of A. When resources are limited to m cells, cells are ranked by their estimated excess: first the hot cells by decreasing $\hat{f}_i$, then the remaining cells by decreasing $\hat{f}_i$. Since y is the standardised logarithm of the counts, $\hat{f}_i$ is measured in standard deviations of $\log(1 + w_i)$: a value of 0.5 means that the cell and its neighbourhood have about half a standard deviation more log sightings than the covariates predict.

All the steps are summarized in Algorithm 1.

Algorithm 1 Covariate-adjusted hot and cold spots

Input: counts $w = (w_1, \ldots, w_n)^\top$, covariates Z, contiguity matrix A; level α, threshold c, permutations B, budget m.
$y \leftarrow$ standardised $\log(1 + w)$; $X \leftarrow (\mathbf{1}, Z)$ with Z standardised; $u \leftarrow M_X y$.
Compute z_0 from (3)–(4). **If** its p-value $> \alpha$, label all cells neutral and **stop**.
$A = E\Delta E^\top$; $\mathcal{C} \leftarrow \{j : \delta_j \geq c\,\delta_1\}$.
Choose λ by ten-fold cross-validation; $S \leftarrow$ (9); $\hat{f} \leftarrow Sy$.
for $b = 1, \ldots, B$ **do**
    $\hat{f}^{(b)} \leftarrow S\,\Pi_b u$ ▷ equivalent to refitting (7) to $X\hat{\beta}_{\mathrm{OLS}} + \Pi_b u$
end for
Label each cell hot, cold or neutral with (10).
Hot (cold) zones $\leftarrow$ connected components of hot (cold) cells in A.
Priority list $\leftarrow$ hot cells by decreasing $\hat{f}_i$, then the other cells by decreasing $\hat{f}_i$; return the first m.

The choices made in each line of Algorithm 1 are justified in Section 3.2.

3 Simulation study

In this section we test the procidure Algorithm 1, identify its characteristics, properties, and also compare it with standard hot-spot methods on data where the true hot and cold areas are

known. All code is available in the project repository [To write: link]. [1]

3.1 Design

Points are generated in the square window $[0, 1000]^2$. Every pattern contains three components:

- 2000 points distributed uniformly (background);
- 3000 points whose intensity follows an environmental covariate: a Gaussian bump $b(u) = \exp\{-\|u - u_0\|^2/(2 \cdot 150^2)\}$ centred at a random location u_0 away from the clusters, with points accepted with probability $\exp\{3(b(u) - 1)\}$;
- clusters, generated as a modified Thomas process: each cluster has a centre and a fixed number of points distributed uniformly within a disc (or an ellipse) around it.

The covariate area is part of the explained variation: the model receives the covariate b evaluated at the centre of each cell, together with an irrelevant covariate drawn from a standard normal distribution. A method that adjusts for covariates should not label the covariate area as hot.

Table 1 lists the scenarios. Scenarios B, L, E, M and S vary the size, shape and number of the clusters; K adds a true cold area, where only a fraction of the points is kept; U has the same data as B but the model does not receive the covariate, to show the effect of a missing covariate; C and N contain no clusters. Every scenario except C and N has a weak and a strong clustering version.

Points are counted on a 20×20 grid (cell side 50) and on a 40×40 grid (cell side 25), with queen contiguity. We generate 100 datasets per setting on the 20×20 grid and 50 on the 40×40 grid, 2400 datasets in total. The permutation band uses $B = 999$ permutations on both grids, as in the application.

The proposed procedure (Algorithm 1, denoted ESF) is compared with:

- Getis-Ord G_i^* (Getis and Ord, 1992; Ord and Getis, 1995) on the residuals u of $\log(1+w)$ on the covariates, with binary weights including the cell itself; a cell is hot if $G_i^* > 1.645$ and cold if $G_i^* < -1.645$. We consider three versions: alone, preceded by the test of Step 1, and preceded by the test with a false discovery rate correction (Benjamini and Hochberg, 1995);

[1] In order to create the link, I need to know if the data can be shared publicly or not. Jorge, can I share it? If not, a summarized version of the data can be uplated which will allow reviewers to see the code without sharing the original data.

Table 1: Simulation scenarios. All patterns also contain 2000 uniform points and 3000 points that follow the covariate; the window is 1000×1000.

Scenario	Clusters	Points per cluster (weak / strong)	Purpose
B	3 discs, radius 100	50 / 200	compact clusters
L	2 discs, radius 200	200 / 600	large, diffuse clusters
E	3 ellipses, semi-axes 300×33	100 / 300	elongated clusters
M	3 discs, radii 50, 100, 200	same intensity: 24, 94, 377 / 50, 200, 800	clusters of different sizes
S	8 discs, radius 40	15 / 40	many small clusters
K	as B (200 points)	cold disc of radius 150 keeps 50% / 20%	a true cold area
U	as B	50 / 200; covariate not given to the model	missing covariate
C	none	–	covariate area only
N	none	3000 uniform points, no covariate area	complete spatial randomness

- local Moran's I (LISA) (Anselin, 1995) on the same residuals, with conditional permutation p-values (999 permutations) and preceded by the test; a cell is hot if it is high–high with $p < 0.05$, cold if low–low with $p < 0.05$;
- two rankings without spatial smoothing: the cells with the largest residuals u (top residuals) and the cells with the largest counts (top counts).

The G_i^* ranking orders cells by G_i^*, and the LISA ranking orders high–high cells by their local I.

A cell belongs to a cluster if its centre lies inside the cluster or if it contains the cluster centre. For a budget of 5% of the cells (20 or 80 cells), we report the *precision* of the priority list, the share of the selected cells that belong to a cluster, and the share of clusters with at least one selected cell. For the labels we report the share of hot cells that belong to a cluster, the share of cluster cells labelled hot, the percentage of the map labelled hot when there are no clusters (C and N), the same two shares for cold cells and the cold area (K), and the share of hot cells inside the covariate area (U).

3.2 Choices within the procedure

The choices listed below rest on three sources of evidence. Choices 1, 6, 7 and 8 were made with preliminary experiments, run before the main study on scenarios A (three clusters of 500 points and 2000 uniform points, no covariate area), B, C and N and on the same two grids (and a 10×10 grid for choice 1). Choices 2, 4 and 5 are supported by the main study, whose results are reported in Section 3.3; for choice 2, the preliminary experiments also applied G_i^* and local Moran's I to the counts. Choice 3 comes from a further analysis of 900 datasets of the main study (scenarios B, S and C), refitted with a natural spline for each covariate. The scripts and full results are in the repository.

Each item below refers to the line of Algorithm 1 and the step of Section 2 that it justifies.

1. **Logarithm of the counts** (Algorithm 1, line 2: $y \leftarrow$ standardised $\log(1 + w)$; Step 1). Counts of sightings are skewed and their variance grows with the mean, so a few cells with many reports dominate a model for the raw counts. On the log scale, the test of Step 1 correctly finds no spatial structure when the pattern is fully explained by a covariate; on the raw scale it does not.

2. **Covariate adjustment** (line 2: $X \leftarrow (\mathbf{1}, Z)$ and $u \leftarrow M_X y$; Step 1). Without covariates, a hot spot is simply an area with many sightings, and areas where the environment favours the pest are flagged even when nothing else is going on. In the simulation study, ranking cells by their counts (or applying Getis-Ord G_i^* and local Moran's I to the counts) placed most of the selected cells in an area driven by a covariate: the share of the top 5% of cells inside a true cluster was at most 0.32, and close to zero with weak clusters.

3. **Linear covariate terms** (line 2: the columns of X; Step 1). A natural spline with 2 to 4 degrees of freedom for each covariate reduced the false rejections of the test in scenario C from 19–32% to 4–6%, but also reduced the rejections with weak small clusters (S) from 64–76% to 28–44%: a flexible function of a smooth covariate absorbs part of the cluster signal (Hodges and Reich, 2010). We keep linear covariate terms and report the spline version as a sensitivity analysis.

4. **Test before mapping** (line 3; Step 1). Local statistics label some cells as hot even when there is no clustering at all: at the 5% level, about one cell in twenty. Testing the residuals first removes most of these false hot cells: in the simulation study, G_i^* labelled 4.0–5.6% of the map hot when there were no clusters, and 0.2–1.9% when it was preceded by the test (Table 3), without any change in its ranking.

The test does not depend on how the cells are labelled afterwards, so it can be used as a first step for any local statistic. In the simulation study we also apply it to the benchmarks (G_i^* and local Moran's I), so that differences between methods in the number of false hot cells come from the local statistic and not from the test.

5. **Eigenvector spatial filtering** (lines 4–5; Step 2). ESF estimates the spatial component inside a regression, so the covariates are adjusted for in the same model, and it represents patterns at several scales at once. Local statistics such as G_i^* average each cell with a fixed neighbourhood. In the simulation study this difference matters when clusters of different sizes coexist: ESF finds the smaller clusters more often.

6. **Smooth candidates** (line 4: $\mathcal{C} \leftarrow \{j : \delta_j \geq c\,\delta_1\}$; Step 2, equation (5)). Eigenvectors with small or negative eigenvalues describe patterns that alternate between neighbours and fit cell-level noise. Restricting the candidates to (5) improved the precision of the top-ranked cells mainly where it was low: with weak clusters and on the finer grid it roughly doubled (for example from 0.17 to 0.36 with cross-validated penalties), with little change elsewhere.

7. **Ridge with a cross-validated penalty** (line 5: choice of λ and of the estimator; Step 2, equation (7)). We compared eight ways to choose the penalty (the Moran's I Lasso rule $|z_0|^{-a}$ with $a = 1, 2, 3, 4$; cross-validation with the smallest error and with the one-standard-error rule; BIC; and the largest penalty whose residuals pass the Moran test), each with all eigenvectors or with the smooth candidates (5). The rule $|z_0|^{-2}$ of the Moran's I Lasso (Barde *et al.*, 2025) has good properties for estimating β (Barde *et al.*, 2024), but for mapping f it ties the amount of smoothing to the strength of the Moran signal: with weak clusters ($|z_0| \approx 3$) the penalty is large and often no eigenvector is selected, so $\hat{f} = 0$; with strong clusters ($|z_0| > 10$) it is so small that almost all eigenvectors enter and $\hat{f}$ follows the noise. With this rule and all eigenvectors, the precision of the top 5% with weak clusters was 0.19 (20×20) and 0.11 (40×40): the Lasso selected no eigenvector in 6 of 17 datasets with residual structure. Cross-validation with smooth candidates raised these values to 0.52 and 0.36 while keeping 0.98–1.00 with strong clusters, and replacing the Lasso by ridge, which keeps all smooth candidates instead of selecting a few, raised them further to 0.67 and 0.47. Our aim is to map f, not to select eigenvectors consistently, so the theoretical reservations about cross-validation for variable selection (Chetverikov *et al.*, 2021) do not apply here. Spatial (block) folds for the cross-validation

did not change the results, which is consistent with the near-uniform shrinkage described after (8). Figure 2 summarises the comparison: rules that penalise little ($|z_0|^{-3}$, $|z_0|^{-4}$) do well with weak clusters but lose precision with strong ones, rules that penalise more ($|z_0|^{-1}$, the one-standard-error rule, BIC) fail with weak clusters, and cross-validation with smooth candidates is the only Lasso rule that does well in both; ridge improves on it with weak clusters.

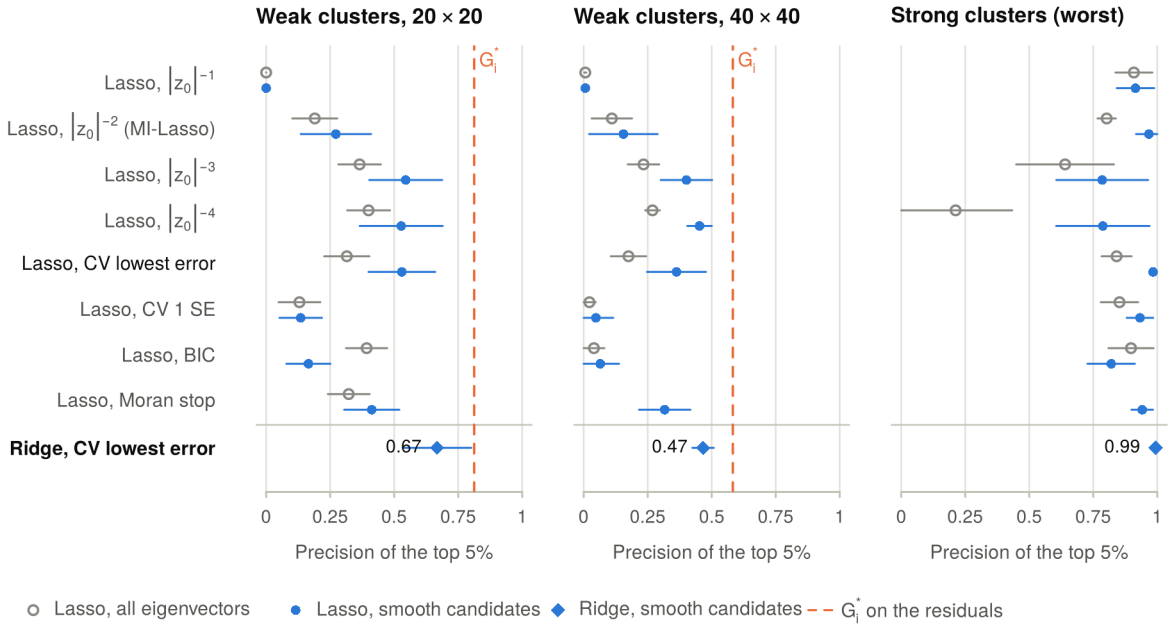


Figure 2: Choice of the estimator and of the penalty in Step 2 (choice 7). Precision of the top 5% of cells for eight penalty rules for the Lasso, each with all eigenvectors (open circles) or with the smooth candidates (5) (filled circles), and for ridge with the cross-validated penalty on the smooth candidates (diamonds). Weak clusters: scenario B with 50 points per cluster. Strong clusters: the lowest mean over scenario A and scenario B with 500 and 200 points per cluster (20×20) and with 200 points (40×40). Means over 20 (20×20) and 10 (40×40) datasets; bars show ± 2 Monte Carlo standard errors. Dashed line: G_i^* on the residuals of the same datasets. The Moran stop rule with all eigenvectors was not run on the 40×40 grid.

8. **Permutation band** (lines 6–9; Step 3, equations (9) and (10)). The distribution of $\hat{f}_i$ under no spatial structure depends on the covariates, the candidate set and the penalty, and has no simple closed form. Permuting the residuals and applying the same smoother S reproduces the estimation in each replicate, so the band accounts for the smoothing that ESF itself introduces. Because ridge is linear in y, this costs one matrix product rather than B refits (a few seconds per dataset on the 40×40 grid with $B = 999$, most of it for the eigendecomposition and the cross-validation). A band for the maximum of

$\hat{f}$ over the map, which would control the chance of any false hot cell, labelled almost no cells in our simulations and was not retained.

Iterative reconstruction. The iterative reconstruction of the spatial graph and the ranking by harmonic centrality are evaluated in Appendix A. They did not improve the precision of the priority list and, without hot/cold labels, about half of the top-ranked cells were cold.

3.3 Simulation results

Table 2 and Figure 3 show the precision of the top 5% of cells. With strong clusters, ESF, G_i^* and LISA all reach 0.99–1.00 in scenarios B, L, M and K. Differences appear with weak clusters. With weak compact clusters (B), G_i^* is the most precise (0.81 and 0.59 on the two grids), followed by LISA (0.75 and 0.52) and ESF (0.70 and 0.43). With clusters of different sizes (M), all methods select cluster cells, but ESF reaches more of the clusters: on the 40×40 grid it includes the smallest cluster (radius 50) in its top 5% in 96% (strong) and 88% (weak) of the datasets, against 68% and 62% for G_i^*; on the 20×20 grid it includes the radius-100 cluster in 97–100% of the datasets, against 79–90% for G_i^*. When clusters are smaller than a cell neighbourhood (S, and E on the coarser grid), ranking the residuals without smoothing is the most precise, since both ESF and G_i^* average over an area larger than the clusters. Ranking by counts fails in every scenario (precision at most 0.32): the covariate area dominates the counts, which shows that covariate adjustment is the most important step for all methods.

Table 3 and Figure 4 summarise the hot and cold labels. With strong compact clusters, 78% of the ESF hot cells belong to a cluster, against 67% for G_i^*; with weak clusters the methods are within a few points of each other, and G_i^* finds slightly more cluster cells. G_i^* with the false discovery rate correction has the most precise labels (0.88–1.00) but finds few cluster cells when clusters are weak (1–15%).

Without the test of Step 1, G_i^* labels 4.0–5.6% of the map hot in scenarios C and N, as expected from its 5% level. With the test, ESF labels 0.2–2.0% hot, G_i^* 0.2–1.9% and LISA 0.1–1.3%. The test therefore provides most of the protection against false hot spots, for all local methods. It rejects in 3% of the datasets with random points on the 20×20 grid, and in 19% (C, 20×20) and 32% (C, 40×40) of the datasets that contain only the covariate area. The rejections in C occur because the covariate enters the model linearly while its effect on $\log(1 + w)$ is not linear; a spline for each covariate reduced them to 4–6%, at the cost of power with weak clusters (Section 3.2). On random points with the 40×40 grid the test rejected in

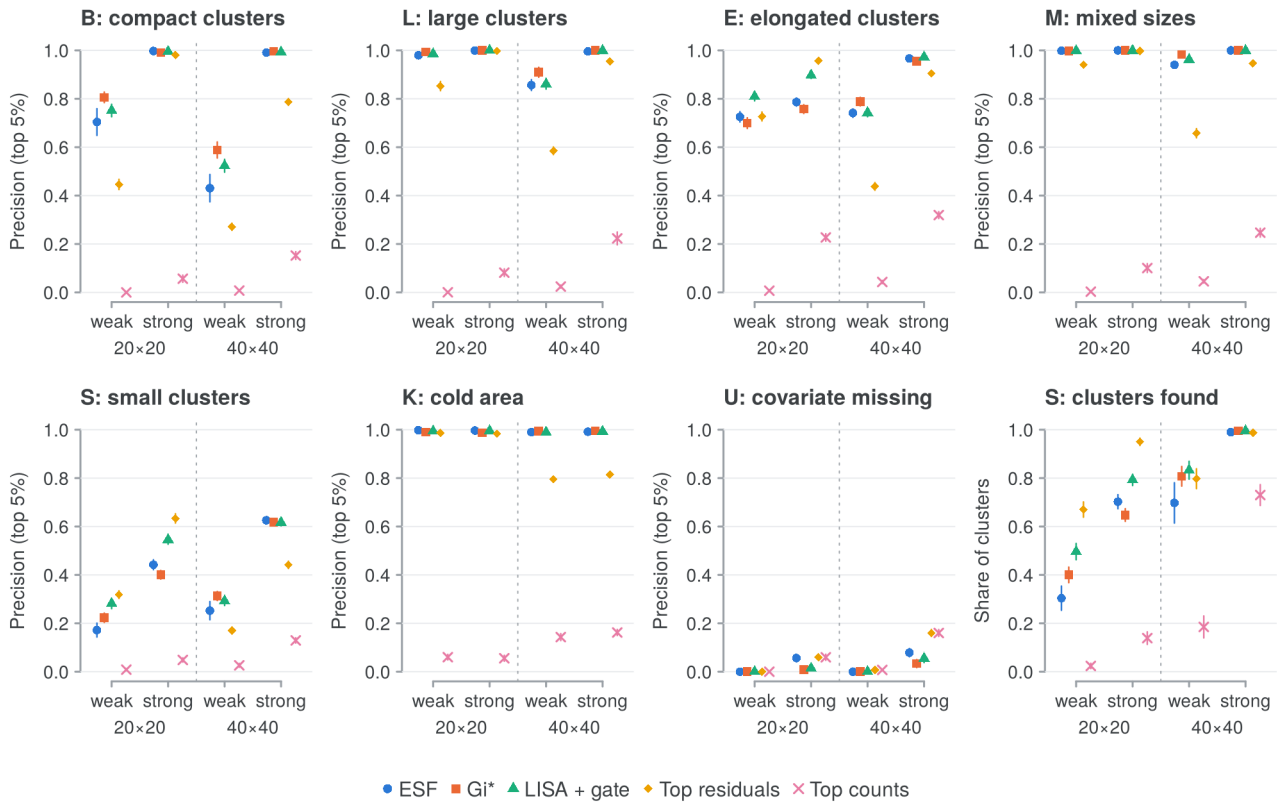


Figure 3: Precision of the top 5% of cells by scenario, grid and cluster strength; bars show ±2 Monte Carlo standard errors. The last panel shows, for scenario S, the share of clusters with at least one cell in the top 5%.

8 of 50 datasets; a check with 500 further datasets gave 5.0% (95% interval 3.3–7.3%), so the test holds its level.

In scenario K with a strong cold area, ESF labels 97% (20×20) and 87% (40×40) of the cold-area cells cold, and 81% and 62% of its cold cells lie in the cold area. G_i^* labels more of the cold area (100% and 96%), but a lower share of its cold cells lies inside it (70% and 58%); LISA has the most precise cold labels among the methods without the correction (81% and 74%). Outside the cold area, cold labels mark cells with few points by chance, which is more frequent on the finer grid.

When the model does not receive the covariate (U), every method fails in the same way: about half of the hot cells (48–64%) and most of the top 5% (82–99%) fall in the covariate area. A hot spot is only defined relative to the covariates that are in the model, which is why the choice of covariates in the application deserves care.

4 Application to cockroach sightings in Madrid

The Technical Unit for Vector Control (TUVC) of the municipality of Madrid work to determine the advance of urban pest in the city, specially rats and cockroachs, for that they activated the comunity in order to determine relevant areas of events. When citizens sight an urban pest they report it to the TUVC, and a member of staff visits the location to confirm the sighting within 24 hours. The data contains the date and location of every confirmed cockroach and rat sighting from January 2010 to December 2013 (Tamayo-Uria *et al.*, 2014). Cockroaches are active in warm months (Ifeanyi and Olawumi, 2015): 87% of the cockroach reports fall between April and September, so we use these months for cockroaches, which gives 1983, 1852, 1960 and 1315 reports in the four summers. Rat reports are spread over the year and are used for all months.

The maps of different summers are very similar (correlation of $\log(1 + w)$ per cell between summers 0.90–0.94). We therefore estimate the main map from the three summers 2010–2012 pooled (5795 reports), use each summer separately to assess persistence, and keep 2013 to evaluate the priority list on data not used to build it.

The study region is the union of the Madrid districts, cut to the bounding box of the sightings plus 1 km. We divide it into square cells of 1 km and drop the cells with less than 0.75% of their area inside the region, which leaves $n = 493$ cells with 1798 queen-contiguity edges. The choice of cell size is examined in Section 4.4.

The covariates come from the database compiled by Tamayo-Uria *et al.* (2013) and used by Tamayo-Uria *et al.* (2014) and Ayyad *et al.* (2018). It records, for each residential address point of the city (about 120 000), the values of environmental, building and census variables at that location. We removed eight columns that were exact copies of other columns or constant, and computed the value of each covariate in a cell as the mean over the address points in it; the 162 cells without any residential address point (a third of the region, mostly non-residential land) take the mean of their neighbours, and their residential count is zero.

We build the model in three nested steps (Table 4):

- **M1, exposure.** Sightings are reports, so they depend on how many people live in a cell and can see and report a pest. M1 contains the logarithm of one plus the number of residential address points in the cell, the population density, and the logarithm of the cell area (cells at the border are smaller).

- **M2, environment.** M1 plus the risk factors for urban pests documented by Tamayo-Uria

et al. (2014) and Ayyad *et al.* (2018): distance to the nearest green area and its size, year of construction of the building, distance to the nearest body of water and to the nearest drinking fountain, distance to the nearest cat-feeding point, and the unemployment rate. M2 is our main model.

- **M3, rat reports.** M2 plus the logarithm of one plus the number of rat reports in the cell during the same years. Cockroach and rat reports per cell are highly correlated (0.93 on the log scale), although the two species have different biology. Much of the shared pattern is likely to reflect where people report rather than where pests are, so M3 uses rat reports as a control for reporting habits. M3 is conservative: it also removes the risk factors that the two species share and that are not in the model, such as the state of the sewers.

The database also contains variables recorded by the TUVC during its field inspections (defective sewers, vacant lots, ruined buildings). Because these inspections may follow the reports, these variables can be a consequence of the response; we do not include them in the main model and use them only as a sensitivity check. Two further covariates of the initial M2, the distance to the nearest market and the mean income, were removed after the relevance check of Section 4.1.

4.1 Choice of covariates

Exposure explains most of the variation: M1 alone accounts for 80.0% of the variance of $\log(1 + w)$ in 2010–2012, M2 for 82.7% and M3 for 87.5%. In every model the residuals keep clear spatial structure (Moran z_0 of 10.8, 9.1 and 5.6), so the test of Step 1 lets the procedure continue.

To decide which environmental covariates to keep, we estimated their coefficients with the spatial structure taken into account: an ordinary least squares fit of $\log(1+w)$ on the covariates and the smooth eigenvectors selected by cross-validated Lasso, so that the eigenvectors absorb the residual autocorrelation (Griffith, 2003). Without the eigenvectors, the standard errors are too small; for example, the p-value of the distance to water drops from 0.24 to 0.0003. We also checked whether each coefficient has the same sign in each of the four summers, and how much the hot cells change when the covariate is dropped.

Figure 5 shows the result for the final M2. Older buildings, larger green areas, closeness to a drinking fountain and higher unemployment are associated with more reports ($p < 0.001$), and the coefficients have the same sign in every summer. The distance to green areas ($p = 0.04$) and

to cat-feeding points ($p = 0.07$) have weaker effects with a stable sign. The distance to water is not related to cockroach reports ($p = 0.24$) but is to rat reports ($p = 0.002$); we keep it so that M2 is the same for both species. In the initial model, the distance to markets and the mean income had p-values of 0.69 and 0.81, their sign changed between summers, and dropping either left at least 91% of the hot cells unchanged; we removed them. Of the exposure variables, the residential address points have by far the largest effect (0.63 standard deviations of $\log(1 + w)$ per standard deviation), followed by density (0.14); the cell area has no effect but is kept as a design variable.

4.2 Hot and cold spots

Figure 6 shows the hot and cold cells of each model for 2010–2012. With M2, 68 cells (14% of the region) are hot and 61 are cold; the hot cells hold 38% of the reports. The hot cells form a band across the south-centre of the city and an area in the north-centre, with smaller zones to the west and north-east. Adding the environmental covariates changes the map: the Jaccard index between the hot cells of M1 and M2 (shared cells divided by cells hot in either) is 0.56, so the hot spots of M1 are partly explained by the environment. Controlling for rat reports (M3) keeps the main zones but reduces them to 59 hot cells (Jaccard 0.53 with M2). The map is robust to the other choices: using the alternative cat variable in the database gives a Jaccard index of 0.90 with M2, and adding the field-campaign variables gives 0.74, while raising the explained variance by only 0.7 points.

The same model applied to rat reports gives a very different map (Jaccard 0.26 with the cockroach M2 map), although the raw counts of the two species are almost the same map. Once exposure and environment are accounted for, the excess of reports is specific to each species, which supports the use of covariate-adjusted hot spots rather than raw counts.

Figure 7 shows the maps of each summer. The test of Step 1 rejects in every summer and model (z_0 between 7.2 and 9.4 for M2, and between 4.6 and 6.2 for M3). With M2, between 54 and 66 cells are hot each summer, and 61–75% of the hot cells of one summer are hot again the next. Twenty-seven cells are hot in all four summers and 70 in at least two; with M3 the numbers are 12 and 60. The cells that are hot every year are the natural first targets for control.

4.3 Evaluation of the priority list

We evaluate the priority lists on the following summer: the cells are ranked with the data of summer t (or of 2010–2012) and the first 25 or 50 cells (5% or 10% of the region) are evaluated

with the reports of summer $t + 1$ (or 2013). We report two measures. The share of the next summer's sightings that fall in the selected cells measures how many reports a control campaign in those cells would address. The share of the next summer's excess measures how much of the reports not explained by the covariates it would address: we fit a Poisson regression of the next summer's counts on the M2 covariates, define the excess of a cell as the observed count minus the fitted count, and report the sum of the excess in the selected cells divided by the sum of the positive excess. We compare ESF with M1, M2 and M3, G_i^* and LISA on the M2 residuals, the largest M2 residuals, the largest counts, and random cells.

Table 5 shows the results. Ranking by counts captures the most sightings (29% with 25 cells, 49% with 50 cells, against 5% and 10% for random cells), because sightings follow exposure, which is stable from year to year. The covariate-adjusted rankings capture fewer sightings (ESF with M2, 17% and 32%) because they deliberately do not target the cells that have many reports only because many people live there. For the excess, the ranking without smoothing (largest residuals, 23–53%) and LISA (23–49%) do best, followed by ranking by counts; ESF with M2 captures 6–27% with 25 cells and 24–40% with 50 cells, and G_i^* 5–31%. Two features of the data explain this. First, the excess is persistent at the level of single cells: the M2 residuals of consecutive summers have correlations of 0.72–0.79, and 10 to 16 of the 25 cells with the largest residuals keep that position the next summer, and a smoothed map spreads the priority over their neighbours, as in scenario S of the simulation study. Second, the excess is measured on the count scale, where the cells with the largest counts also have the largest deviations from the fitted values. For a control campaign this suggests using the hot zones of ESF to decide where to act and the residuals within each zone to order the cells, as in Step 4.

4.4 Cell size

The results of an analysis on areal units depend on the size of the units (Openshaw, 1984). We repeated the analysis of 2010–2012 with cells of 500 m and 2 km (Table 6, Figure 8). To compare maps on different grids, we overlay a common lattice of 250 m cells and call a 250 m cell hot for a grid if its centre lies in a hot cell of that grid.

The test of Step 1 rejects for every cell size and model, although with 2 km cells the evidence under M3 is weak ($z_0 = 2.1$). Larger cells label a larger share of the city hot (9%, 14% and 22% with M2). The main zones, the south-central band and the north-central area, appear at all three sizes, but their extent changes: the Jaccard index between the hot areas is 0.29 for 500 m and 1 km, 0.35 for 1 km and 2 km, and 0.25 for 500 m and 2 km. The cell size should

therefore be chosen with the intervention in mind: a 1 km cell is a manageable unit for a control campaign in a city of this size, and finer cells locate the excess more precisely but contain fewer reports each (65% of the 500 m cells have none).

5 Discussion

We have proposed a procedure to map covariate-adjusted hot and cold spots of urban pest reports and to turn the map into a priority list. It combines a test for residual spatial structure, an estimate of the spatial component by eigenvector spatial filtering with smooth eigenvectors and a ridge penalty, a permutation band that labels each cell, and a ranking of the hot cells. We compared it with G_i^*, local Moran's I and rankings without spatial smoothing in a simulation study, and applied it to cockroach sightings in Madrid.

The simulation study shows trade-offs rather than a single winner. With strong clusters, ESF, G_i^* and local Moran's I are equally precise. With weak compact clusters, G_i^* is the most precise, because its fixed neighbourhood matches the cluster size. When clusters of different sizes coexist, ESF reaches the small clusters more often, because the eigenvectors represent patterns at several scales. When the clusters are smaller than a neighbourhood of cells, ranking the residuals without smoothing is the most precise. The choice therefore depends on what the user expects the clusters to look like. We keep ESF as the main method because it adjusts the covariates and the spatial component in one model, so the covariate coefficients are corrected for residual spatial correlation (Section 4.1); it gives a continuous map of the excess with a band for every cell; and it detects hot and cold areas at several scales without fixing a neighbourhood. Local statistics on the residuals are a useful complement and are cheap to compute, and we recommend reporting them alongside.

The step that matters most for error control is the test of Step 1, not the choice of local statistic. Without it, G_i^* labels about 5% of a map without clusters as hot, as expected from its level; with it, ESF, G_i^* and local Moran's I label at most 2.0% of such maps. The test can be placed in front of any local statistic. A false discovery rate correction of the local tests (Benjamini and Hochberg, 1995) gives the most precise labels but misses most cells of weak clusters, which for a control campaign is usually the more costly error.

A hot spot is only defined relative to the covariates in the model. In the simulation, when the environmental covariate was not given to the model, every method placed most of its priority list in the area explained by that covariate. In Madrid, exposure alone explains 80% of the variation in log reports, and adding environmental risk factors and rat reports changes

which cells are hot. Applied to rats with the same covariates, the procedure gives a map that shares little with the cockroach map, although the raw counts of the two species are almost the same map. Practitioners should choose the covariates according to the question: M1 identifies areas with more reports than their population predicts, M2 areas with more reports than their environment also predicts, and M3 areas where cockroaches are reported more than rats.

Evaluated on the following summer, ranking by counts captures the most reports, because reports follow exposure and exposure is stable. Covariate-adjusted rankings target a different quantity, the excess over what the covariates predict. For this target, the residuals of single cells were more persistent than the smoothed map, and ranking them captured more of the next summer's excess than ESF. This agrees with scenario S of the simulation study and suggests a two-level use of the output: the hot zones of the procedure indicate where to act, and the residuals within each zone indicate the order.

Reports measure sightings by citizens, not pest abundance; the exposure variables and rat reports reduce, but do not remove, the influence of reporting habits, and M3 also removes the risk factors that rats and cockroaches share. The covariates were compiled once and are used for all four summers, and a third of the 1 km cells contain no residential address point, so their covariates are taken from their neighbours. The variables recorded during the field inspections of the vector control unit were excluded from the main model because they may follow the reports. The test of Step 1 assumes that the covariate effects are linear on the log scale; when they are not, the residuals keep part of the covariate pattern and the test rejects more often than its level (19–32% of datasets in scenario C). A spline for each covariate removes this, at the cost of power with weak clusters, and can be used as a sensitivity analysis. The results depend on the cell size (Openshaw, 1984): in Madrid the main zones appear with cells of 500 m, 1 km and 2 km, but their extent changes, so the cell size should be chosen with the intervention in mind. Finally, the procedure models $\log(1 + w)$ with a Gaussian approximation, which is adequate for the counts in our application but not for very sparse counts.

The ridge penalty on the smooth eigenvectors is close to the random-effects formulation of ESF (Murakami and Griffith, 2015), which gives each eigenvector a variance that depends on its eigenvalue; that formulation, and a Poisson likelihood for the counts, are natural extensions. The persistence of the hot cells across summers suggests a space–time version, in which the spatial component evolves over time and the priority list is updated as new reports arrive. Because the procedure needs only counts per area, covariates and a contiguity matrix, it can be applied to other allocation problems in which resources must be placed where events exceed what local characteristics predict.

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Author contributions

[To write: update]

Use of artificial intelligence

During the revision, the authors used Claude (Anthropic), a large language model used through an agentic coding and writing environment, as a research assistant. Under the authors' direction, Claude: reorganised the analysis code into an R package and wrote unit tests; implemented the proposed procedure and the benchmark methods (G_i^*, local Moran's I and the rankings without smoothing); proposed and ran, after discussion with the authors, the simulation experiments of Section 3; prepared the Madrid data (grid, exposure and covariates), ran the analyses of Section 4, and checked the covariate definitions against the cited sources; produced the figures and tables; and drafted the text of the Methods and Simulation sections and of Appendix A. All modelling decisions (choice of method, penalty, test, scenarios, data and covariates) were taken by the authors. The authors checked the code, verified the reported numbers against the outputs, and reviewed and edited all text. The authors take full responsibility for the content of this article. Claude is not an author.

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Appendices

A Iterative reconstruction of the spatial weights matrix

This appendix describes an extension of the eigenvector spatial filtering (ESF) step that reconstructs a sparser spatial weights matrix from the selected eigenvectors and iterates the selection on the reconstructed graph. We evaluate it in simulation and explain why it is not part of the proposed procedure.

A.1 The iterative reconstruction

Let A be the $n \times n$ binary contiguity matrix of the study region, Y the standardised counts per cell and Z the standardised covariates. The procedure starts from $D^{(0)} = A$ and, at iteration k:

1. computes the eigendecomposition $D^{(k)} = E\,\Delta\,E^{\top}$, with $\Delta = \mathrm{diag}(\delta_1, \ldots, \delta_n)$, and selects a subset S of eigenvectors with the MI-Lasso estimator, with penalty $\lambda = |z_0|^{-2}$ fixed from the standardised Moran's I of the residuals of Y on Z;

2. reconstructs
$$B = E_S\,\Delta_S\,E_S^{\top}, \qquad b_{ij} = \sum_{s \in S} \delta_s\, e_s(i)\, e_s(j); \tag{11}$$

3. applies a rounding rule: among the edges of $D^{(k)}$ with $b_{ij} > 0$, it keeps those with $b_{ij} \geq \tau^{(k)}$, where $\tau^{(k)}$ is a threshold computed from these values, giving $D^{(k+1)} = D^{(k)} \circ \mathbf{1}\{B \geq \tau^{(k)}\}$;

4. accepts $D^{(k+1)}$ if the post-Lasso residuals obtained with its eigenvectors show no spatial correlation (Moran's I with A, level 0.05), and stops otherwise or when $D^{(k+1)} = D^{(k)}$.

Because each accepted step removes at least one edge, the procedure stops after at most as many iterations as A has edges. Cells are then ranked by a ranking rule applied to the final graph.

We call *baseline configuration* the procedure with all n eigenvectors as candidates, eigenvalue weights in (11), the 75% quantile of the positive b_{ij} as threshold, and cells ranked by harmonic centrality on the final graph. Each variant below changes one element of the baseline.

Rounding rules. *None* (all edges kept, $D = A$; no reconstruction); *quantile* q ($\tau = q$-quantile of the positive b_{ij} on the current edges, $q \in \{0.50, 0.75, 0.80, 0.90\}$); *mixture* (a two-component Gaussian mixture fitted to the positive b_{ij}; τ is the point between the two means where the weighted component densities are equal, and all edges are kept when the Bayesian information criterion selects one component); *permutation* (30 Freedman–Lane permutations of the residuals of Y on Z; for each, the selection and (11) are recomputed on the current graph, and τ is the 95th percentile of the pooled positive b_{ij}).

Ranking rules. Let $\hat{f} = E_S \hat{\gamma}_S$ be the fitted spatial component (centred), and call a cell *hot* if $\hat{f}_i > 0$ and *cold* otherwise. *Centrality, whole graph*: harmonic centrality on the final graph, over all cells (baseline). *Centrality, hot cells*: harmonic centrality computed among hot cells only. *Largest excess*: hot cells ordered by $\hat{f}_i$. *Hot zones first*: connected groups of hot cells ordered by $\sum_i \hat{f}_i$; within a group, cells ordered by harmonic centrality. Two benchmarks that do not use the graph are also reported: *top counts* (cells ordered by Y) and *top residuals* (cells ordered by the residuals of Y, or of $\log(Y+1)$, on Z).

A.2 Simulation design

All scenarios use a 1000×1000 window divided into a regular grid with queen contiguity. Table 7 lists the scenarios.

Unless stated otherwise the grid has 20×20 cells (cell side 50) and the budget is the top 20 cells (5%). For scenarios A and B, *precision* is the share of the selected cells inside a true cluster disc. For scenarios D and D0, edges are classified as kept or removed, and we report the area under the ROC curve (AUC) of b_{ij} for separating real from barrier edges, the share of edges kept, and Youden's J = (share of barrier edges removed) + (share of real edges kept) $- 1$, which is 0 for a classification unrelated to the barriers.

A.3 Results

Sign of the reconstructed weights. Equation (11) is invariant to the sign of each eigenvector, since $e_s e_s^\top = (-e_s)(-e_s)^\top$. A positive b_{ij} indicates that cells i and j lie on the same side of the selected spatial patterns, whether both have high or both have low counts. Under the baseline configuration, half of the 20 cells ranked highest in scenario B had $\hat{f}_i < 0$, and precision was 0.28 in scenario A and 0.24 in scenario B (Fig. 9). Ranking hot cells by $\hat{f}$ gave 1.00 and 0.96. Figure 10 shows the successive graphs and the selected cells for one simulated

map.

Effect of rounding and iteration on the selected cells. We compared rounding with the 75% quantile against no rounding over 162 settings of scenario B: grids of 10×10, 20×20 and 40×40 cells, 50, 200 or 500 points per cluster, penalty exponents $a \in \{0.5, 1, 2\}$ in $\lambda = |z_0|^{-a}$, counts on the original or the $\log(Y+1)$ scale, and three ranking rules (10 replicates per setting). Iteration removed edges in 495 of 1620 runs (31%); in the remaining runs the first reduced graph was rejected by the Moran test. In the runs where edges were removed, the change in precision was +0.06 (95% CI 0.02 to 0.10) for centrality on the whole graph, −0.01 (−0.03 to 0.01) for largest excess and −0.06 (−0.11 to −0.01) for hot zones first (Fig. 11).

Rounding rules. Table 8 and Fig. 12 compare the seven rounding rules with all other elements fixed at the baseline (scenarios A and B, 20 replicates each; scenarios D and D0, 20 replicates each). With the largest-excess ranking, precision was 1.00 in scenario A and 0.95–0.96 in scenario B for every rule, including no rounding. With hot zones first, the 80% and 90% quantiles and the permutation rule gave about 0.10 more than no rounding, and remained below the largest-excess ranking. With the baseline ranking, precision ranged from 0.10 to 0.37. In scenario D0 every rule except the mixture removed between 90% and 97% of the edges.

Fine grid with weak clusters. On a 40×40 grid (cell side 25) with 50 or 200 points per cluster, scenario B, budget of 80 cells, 10 replicates, the rankings that use hot cells and connectivity reached, with no rounding, a precision of 0.44–0.48 (50 points) and 0.81–0.89 (200 points), against 0.29 and 0.80 for top residuals on the log scale. Rounding reduced the precision of hot zones first by 0.24 and 0.25 (75% quantile), 0.18 and 0.17 (90% quantile) and 0.06 and 0.10 (permutation) (Fig. 13).

Recovery of a known graph. In scenario D, the AUC of b_{ij} for separating real from barrier edges was 0.55, 0.55 and 0.45 for $\rho = 0.5$, 0.8 and 0.95 with eigenvalue weights, and 0.55, 0.53 and 0.61 with squared coefficients $\hat{\gamma}_s^2$ as weights. The absolute difference between the residuals of neighbouring cells, $|e_i - e_j|$, gave 0.55, 0.65 and 0.96 (Fig. 14). After the full procedure, Youden's J ranged from −0.02 to 0.00 across rounding rules. With 130–300 of 400 eigenvectors selected, B differs from A by a small number of patterns with global support, while a barrier is a local discontinuity.

A.4 Conclusion

In these simulations, no rounding rule increased the highest precision obtained without rounding by more than 0.01 (largest excess on the 20×20 grid; centrality among hot cells on the 40×40 grid). The reconstructed weights did not identify the edges removed by barriers, and most rounding rules removed most edges when no barrier was present. Ranking by centrality on the whole reconstructed graph selected cold cells in about half of the positions. The proposed procedure therefore uses the ESF step once, to estimate the spatial component $\hat{f}$ conditional on the covariates, and derives hot and cold zones from it on the contiguity graph A.

Table 2: Precision of the top 5% of cells (share of the selected cells that belong to a cluster), mean over replicates. Bold: highest value in the row. The three G_i^* versions share the same ranking.

Scenario	Grid / strength	ESF	Gi*	LISA + gate	Top residuals	Top counts
B	20×20 weak	0.70	**0.81**	0.75	0.45	0.00
	20×20 strong	**1.00**	0.99	0.99	0.98	0.06
	40×40 weak	0.43	**0.59**	0.52	0.27	0.01
	40×40 strong	0.99	**1.00**	0.99	0.79	0.15
L	20×20 weak	0.98	**0.99**	0.99	0.85	0.00
	20×20 strong	1.00	**1.00**	**1.00**	1.00	0.08
	40×40 weak	0.86	**0.91**	0.86	0.59	0.02
	40×40 strong	1.00	**1.00**	1.00	0.95	0.22
E	20×20 weak	0.73	0.70	**0.81**	0.73	0.01
	20×20 strong	0.79	0.76	0.90	**0.96**	0.23
	40×40 weak	0.74	**0.79**	0.74	0.44	0.04
	40×40 strong	0.97	0.96	**0.97**	0.91	0.32
M	20×20 weak	**1.00**	1.00	**1.00**	0.94	0.00
	20×20 strong	**1.00**	**1.00**	1.00	1.00	0.10
	40×40 weak	0.94	**0.98**	0.96	0.66	0.05
	40×40 strong	**1.00**	**1.00**	1.00	0.95	0.25
S	20×20 weak	0.17	0.22	0.28	**0.32**	0.01
	20×20 strong	0.44	0.40	0.55	**0.63**	0.05
	40×40 weak	0.25	**0.31**	0.29	0.17	0.03
	40×40 strong	**0.63**	0.62	0.62	0.44	0.13
K	20×20 weak	**1.00**	0.99	0.99	0.99	0.06
	20×20 strong	**1.00**	0.99	1.00	0.98	0.06
	40×40 weak	0.99	**0.99**	0.99	0.80	0.14
	40×40 strong	0.99	**1.00**	0.99	0.81	0.16
U	20×20 weak	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
	20×20 strong	0.06	0.01	0.01	**0.06**	**0.06**
	40×40 weak	0.00	0.00	0.00	**0.01**	**0.01**
	40×40 strong	0.08	0.03	0.06	**0.16**	**0.16**

Table 3: Hot and cold labels, mean over replicates. Scenario U: lower is better.

Measure	Setting	ESF	Gi*	Gi* + gate	Gi* + gate + FDR	LISA + gate
Hot cells inside a cluster	B strong 20×20	0.78	0.67	0.67	0.88	0.76
	B weak 20×20	0.60	0.63	0.62	0.97	0.68
	M strong 40×40	0.93	0.91	0.91	0.99	0.94
	S weak 40×40	0.28	0.28	0.29	1.00	0.31
Cluster cells labelled hot	B strong 20×20	0.95	0.98	0.98	0.88	0.96
	B weak 20×20	0.54	0.63	0.58	0.15	0.46
	M strong 40×40	0.87	0.92	0.92	0.79	0.84
	S weak 40×40	0.35	0.44	0.35	0.01	0.28
% of cells hot, no clusters	C 20×20	1.4	5.6	1.2	0.0	0.8
	N 20×20	0.2	4.0	0.2	0.0	0.1
	C 40×40	2.0	5.3	1.9	0.0	1.3
	N 40×40	1.0	4.6	0.8	0.0	0.5
Cold cells inside the cold area	K strong 20×20	0.81	0.70	0.70	0.88	0.81
	K strong 40×40	0.62	0.58	0.58	0.93	0.74
Cold-area cells labelled cold	K strong 20×20	0.97	1.00	1.00	0.96	0.98
	K strong 40×40	0.87	0.96	0.96	0.80	0.85
Hot cells in the covariate area	U strong 20×20	0.56	0.48	0.48	0.63	0.53
	U strong 40×40	0.56	0.51	0.51	0.64	0.55

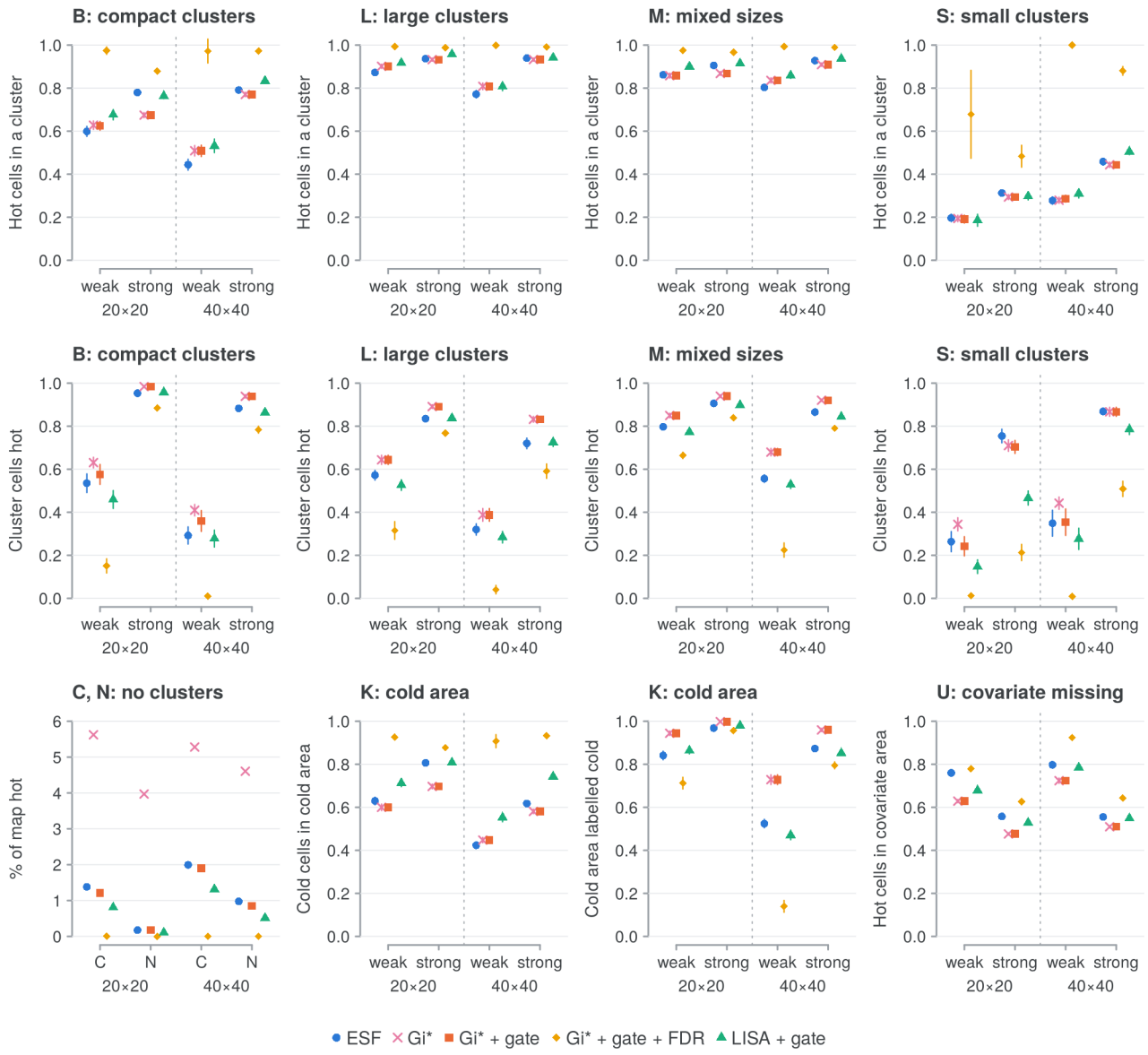


Figure 4: Hot and cold labels by scenario: share of hot cells inside a cluster, share of cluster cells labelled hot, percentage of the map labelled hot without clusters (C, N), cold labels in scenario K, and hot cells in the covariate area when the covariate is missing (U).

Table 4: Covariates of the nested models. Distances are to the nearest feature, in metres, measured from each address point.

Model	Name	Definition	Source
M1	log residential	$\log(1 + \text{residential address points in the cell})$	address database (Tamayo-Uria *et al.*, 2013)
	Densidad	population density [units]	census (INE)
	log area	logarithm of the cell area within the region	grid
M2	dist_verd	distance to a green area	municipality
	AreaVer	surface of the nearest green area	municipality
	YCons	year of construction of the building	land registry
	dist_agua	distance to a body of water	Ayyad *et al.* (2018)
	dist_fuent	distance to a drinking fountain	Ayyad *et al.* (2018)
	D_GatoOK	distance to a cat-feeding point [confirm]	municipality
	Desempleo	unemployment rate (%)	census
M3	log rat reports	$\log(1 + \text{rat reports in the cell, same years})$	TUVC

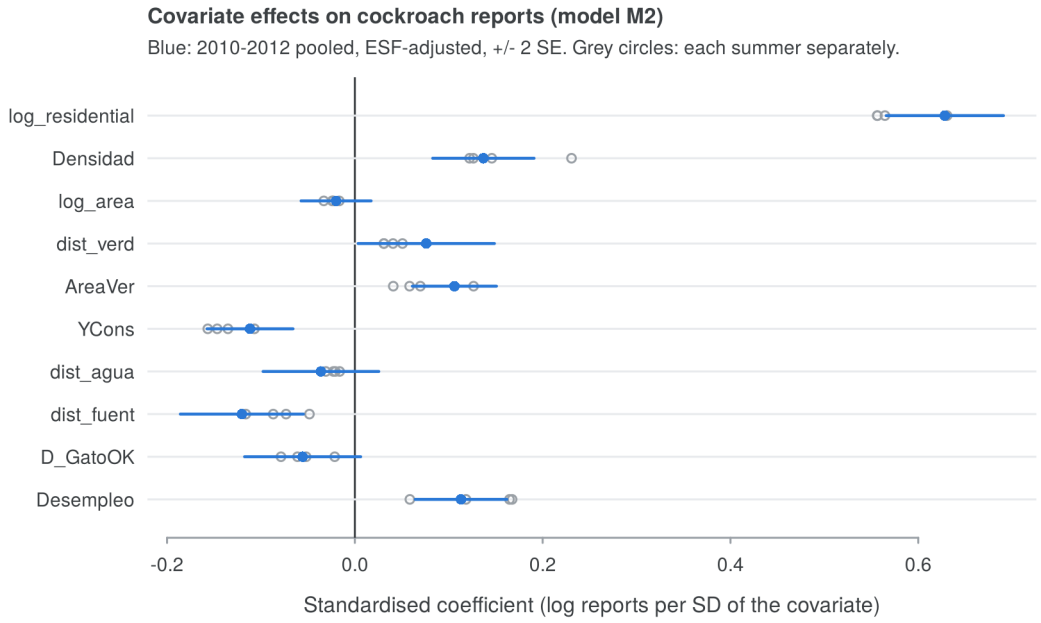


Figure 5: Coefficients of model M2 for cockroach reports, 2010–2012, with the residual spatial structure absorbed by smooth eigenvectors (± 2 standard errors); grey circles are the coefficients estimated in each summer separately. All variables are standardised.

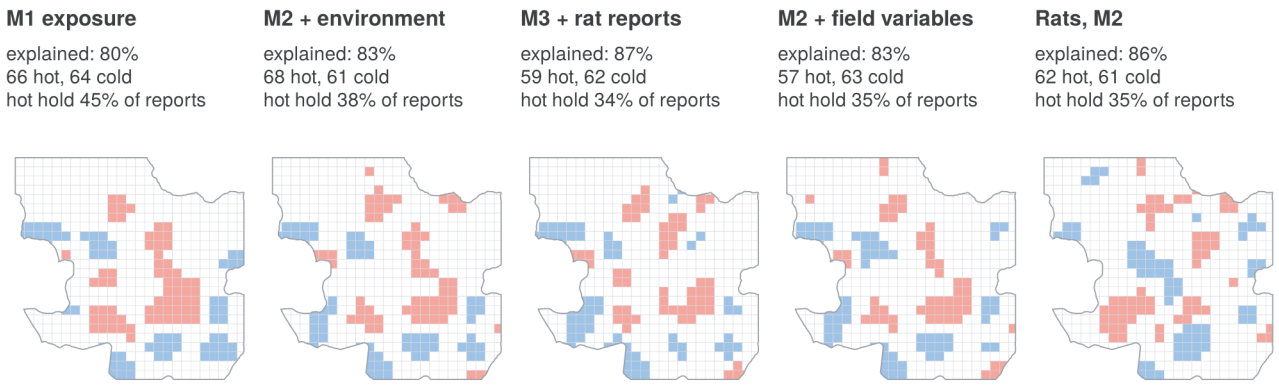


Figure 6: Hot (red) and cold (blue) cells for cockroach reports, April–September 2010–2012, 1 km cells, by model; the last panel applies M2 to rat reports.

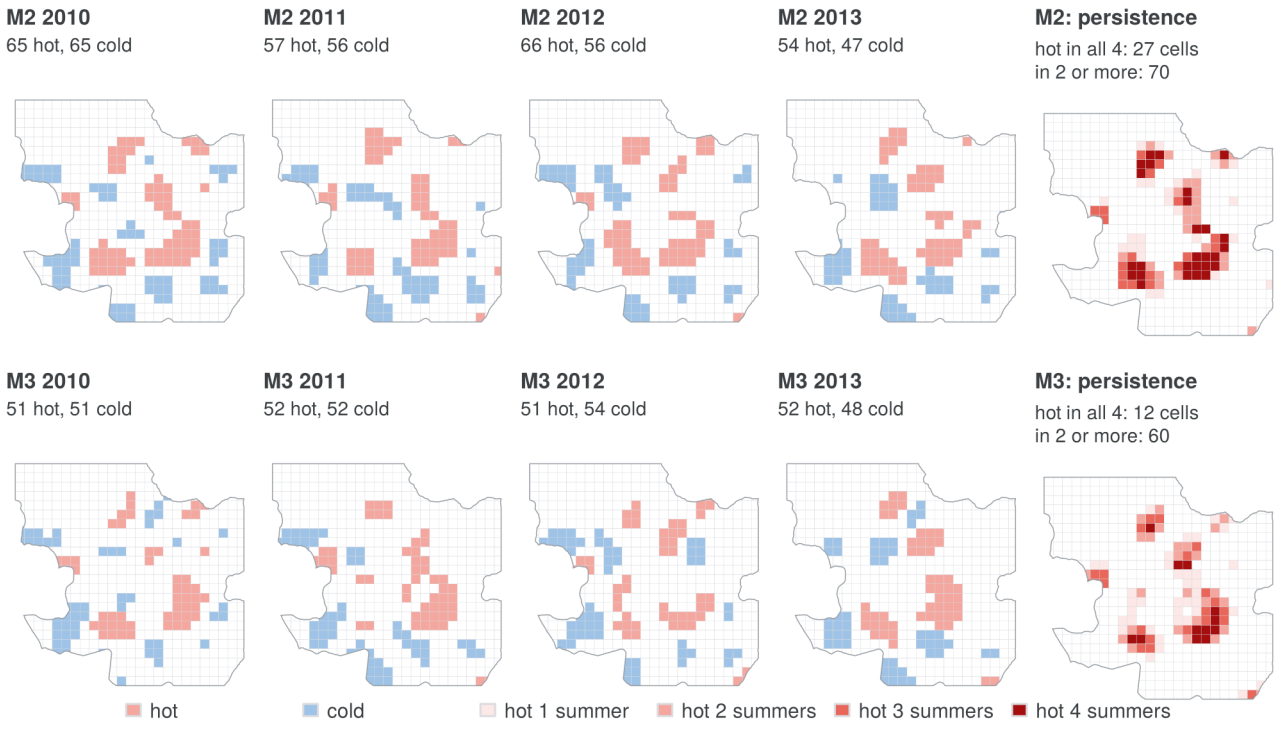


Figure 7: Hot and cold cells in each summer, 2010–2013, for models M2 (top) and M3 (bottom), and number of summers in which each cell is hot (right).

Table 5: Share of the next summer's sightings and of its excess over the M2 covariates captured by the first 25 and 50 cells of each priority list. Each column is a pair of training and test summers.

	Sightings captured (%)				Excess captured (%)			
Ranking	2010→2011	2011→2012	2012→2013	2010-2012→2013	2010→2011	2011→2012	2012→2013	2010-2012→2013
25 cells (5% of the city)								
ESF, M1 exposure	22.4	18.4	21.7	19.2	27.6	13.6	14.3	18.6
ESF, M2 environment	20.4	18.8	16.7	13.2	27.0	21.8	6.0	11.2
ESF, M3 rat reports	19.0	18.3	13.2	11.0	25.9	26.4	7.3	11.7
Gi* (M2 residuals)	19.3	18.2	17.2	12.3	17.5	18.0	7.4	4.6
LISA (M2 residuals)	24.2	23.6	20.0	16.2	37.7	33.1	23.3	23.9
Top residuals (M2)	26.1	23.3	20.1	16.1	45.3	38.4	22.6	24.3
Top counts	29.9	29.3	28.4	28.8	38.2	35.2	26.2	27.1
Random cells	5.1	5.1	5.1	5.1	0.0	0.0	0.0	0.0
50 cells (10% of the city)								
ESF, M1 exposure	38.9	34.5	36.5	33.2	29.3	28.9	26.4	21.3
ESF, M2 environment	35.5	31.1	33.6	28.7	40.4	27.7	28.5	23.9
ESF, M3 rat reports	30.3	31.8	24.9	25.4	31.9	33.7	17.3	26.7
Gi* (M2 residuals)	34.4	31.5	29.0	26.2	31.1	27.0	18.8	18.0
LISA (M2 residuals)	37.9	36.4	33.8	31.0	48.8	41.2	34.5	37.8
Top residuals (M2)	40.7	41.9	37.6	33.2	50.2	52.8	41.4	48.2
Top counts	48.5	50.9	47.8	49.8	41.4	52.3	34.7	42.6
Random cells	10.1	10.1	10.1	10.1	0.0	0.0	0.0	0.0

Table 6: Effect of the cell size, cockroach reports 2010–2012. Share of the city hot and Jaccard index computed on a common 250 m lattice.

Cell size	Cells	Cells without reports	z_0 (M1 / M2 / M3)	Hot cells (M2)	City hot (M2)	Reports in hot cells (M2)
500 m	1901	65%	21.8 / 18.9 / 14.7	171	9%	42%
1 km	493	53%	10.8 / 9.1 / 5.6	68	14%	38%
2 km	136	39%	4.6 / 4.0 / 2.1	25	22%	55%

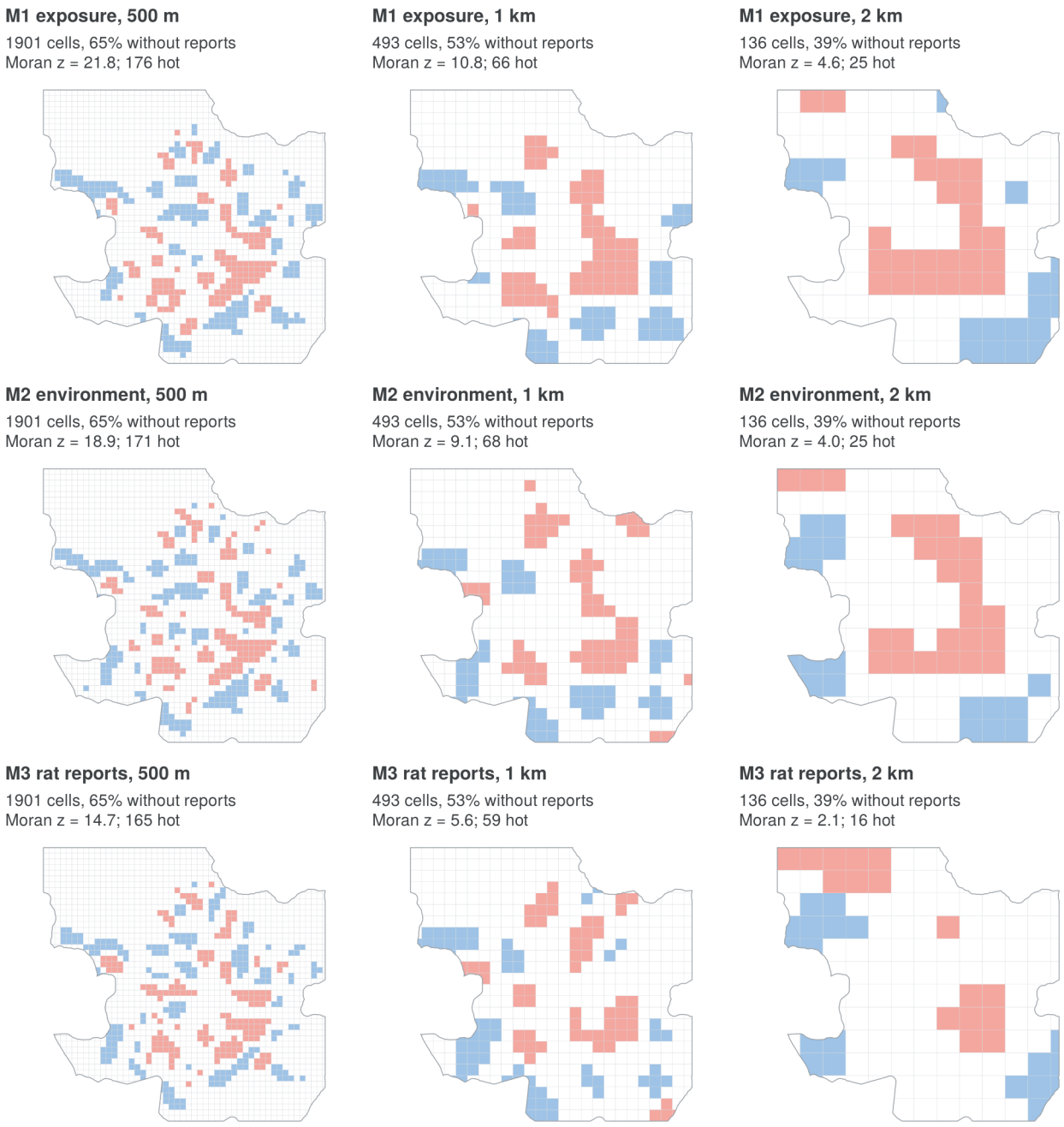


Figure 8: Hot and cold cells for cockroach reports, 2010–2012, with cells of 500 m, 1 km and 2 km (columns) and models M1–M3 (rows).

Table 7: Simulated scenarios.

Scenario	Data-generating process	Target
A: clusters	Three clusters: parents uniform in the window at least 100 units from the border; 500 offspring per parent, uniform in a disc of radius 100; 2000 additional uniform points. No covariate.	Cells inside the three discs
B: clusters and covariate	Scenario A plus 3000 points accepted with probability $\exp\{3(b(x) - 1)\}$, where b is a Gaussian surface (standard deviation 150) centred at least 300 units from every parent. The model receives $Z = (b,\ \varepsilon)$ at the cell centres, with $\varepsilon \sim N(0, 1)$.	Cells inside the three discs, not the covariate area
D: barriers	$Y = \rho WY + Z\beta + e$ on a 20×20 grid, with W the row-standardised version of a graph $W^* \subset A$ in which a vertical line (one crossing left open) and a horizontal line on the left half remove 85 of the 1482 edges; Z a linear gradient and noise, $\beta = (1, 0)$, $\rho \in \{0.5, 0.8, 0.95\}$.	The 85 removed edges
D0: no barriers	As D with $W^* = A$.	No edge removed

Table 8: Precision of the top 20 cells (scenario A / scenario B) by rounding rule and ranking rule, and share of edges kept in scenario D0. Mean of 20 replicates.

Rounding rule	Largest excess	Hot zones first	Centrality, whole graph	Edges kept (D0)
None	1.00 / 0.95	0.85 / 0.76	0.13 / 0.10	100%
50% quantile	1.00 / 0.95	0.81 / 0.71	0.32 / 0.37	6%
75% quantile (baseline)	1.00 / 0.95	0.84 / 0.84	0.22 / 0.29	8%
80% quantile	1.00 / 0.95	0.89 / 0.90	0.19 / 0.31	3%
90% quantile	1.00 / 0.95	0.91 / 0.91	0.26 / 0.32	10%
Mixture	1.00 / 0.96	0.80 / 0.78	0.26 / 0.31	76%
Permutation	1.00 / 0.95	0.90 / 0.91	0.23 / 0.37	10%

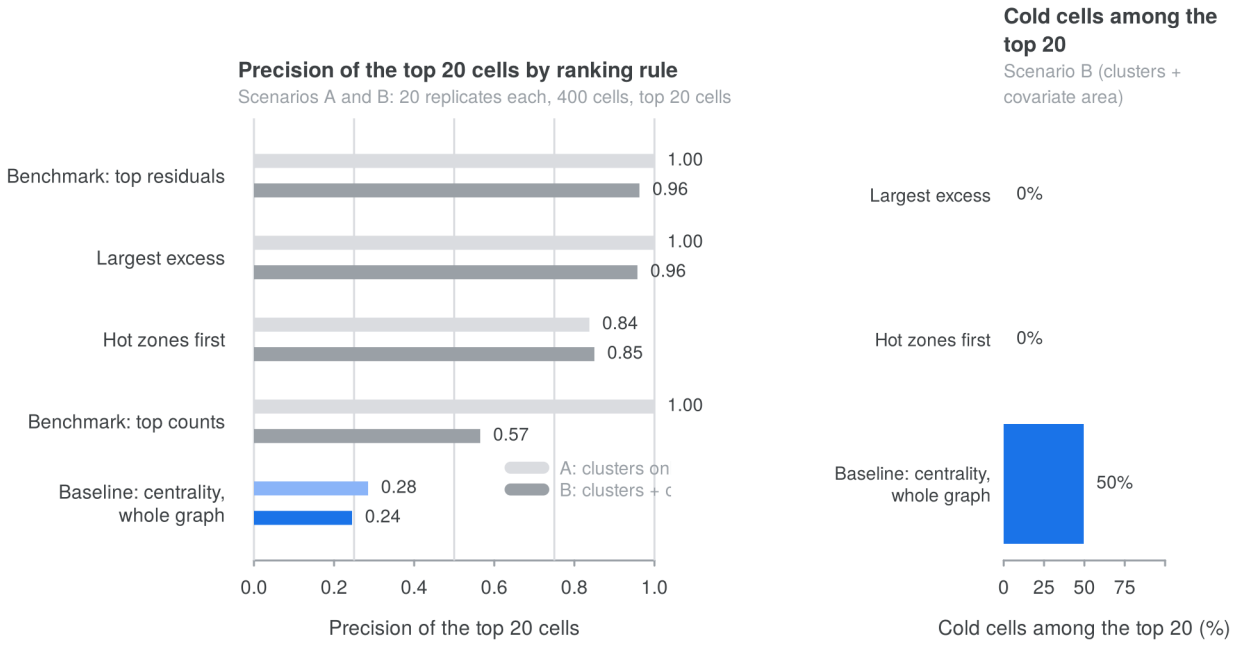


Figure 9: Precision of the top 20 of 400 cells by ranking rule (scenarios A and B, 20 replicates each) and share of cold cells among them (scenario B). Baseline: harmonic centrality on the final graph.

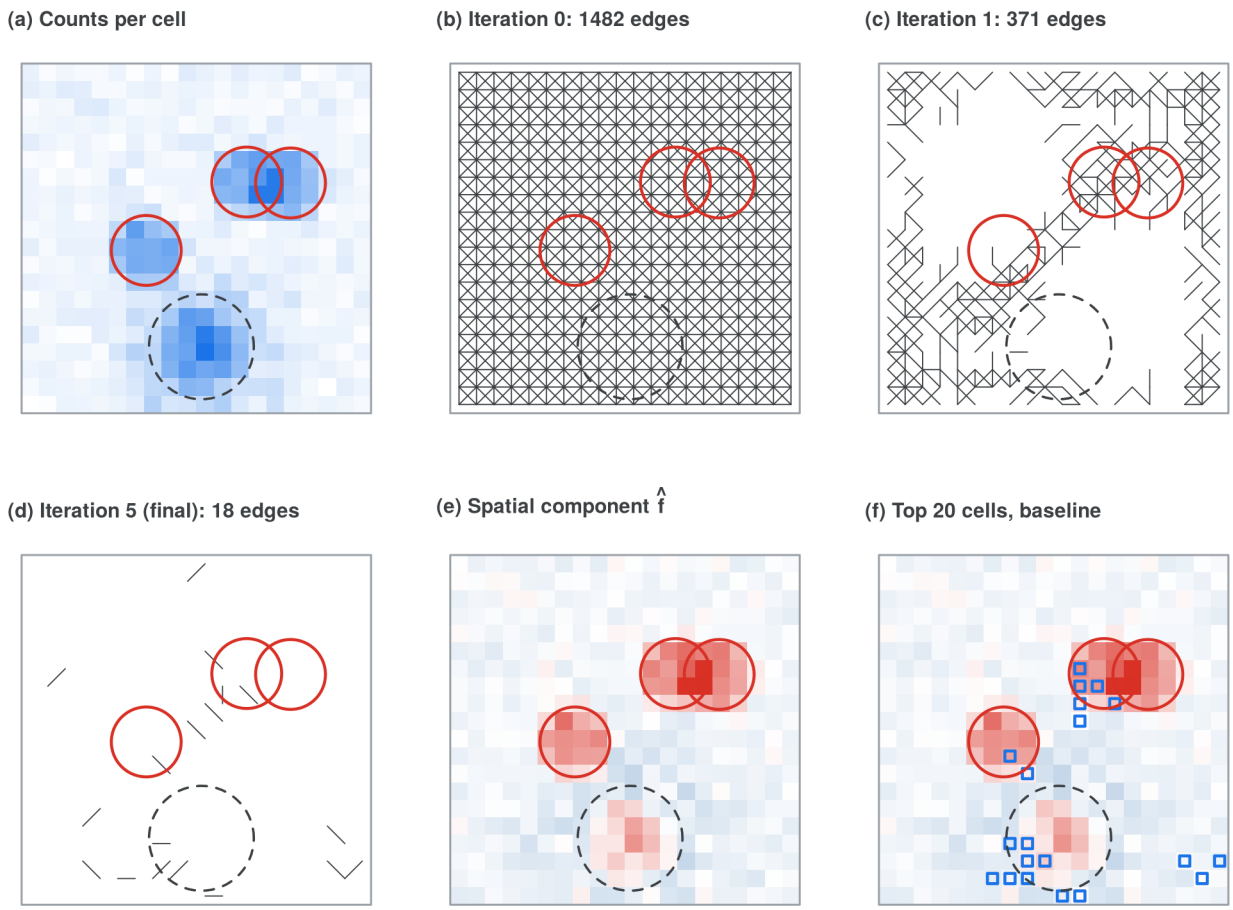


Figure 10: One simulated map from scenario B under the baseline configuration. (a) Counts per cell. (b)–(d) Graph at iterations 0, 1 and at the last accepted iteration. (e) Fitted spatial component $\hat{f}$. (f) Top 20 cells under the baseline ranking. Circles: true clusters; dashed circle: covariate area.

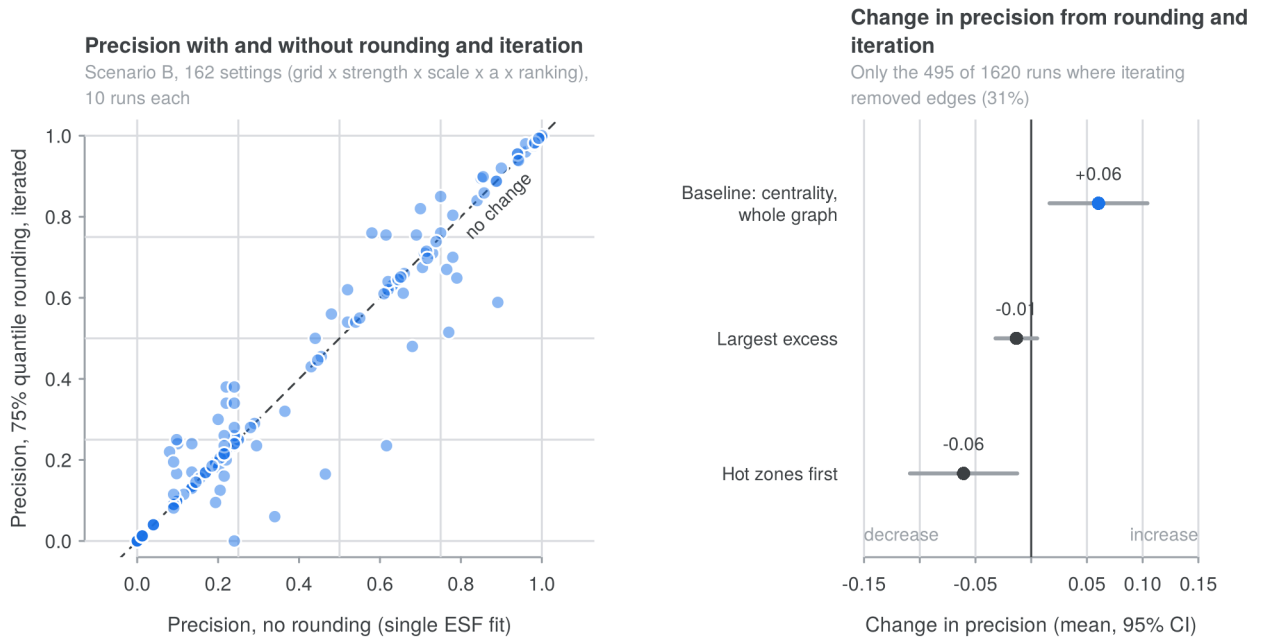


Figure 11: Left: precision with no rounding against precision with 75% quantile rounding and iteration; each point is one setting (mean of 10 replicates). Right: change in precision in the runs where iteration removed edges (mean and 95% confidence interval).

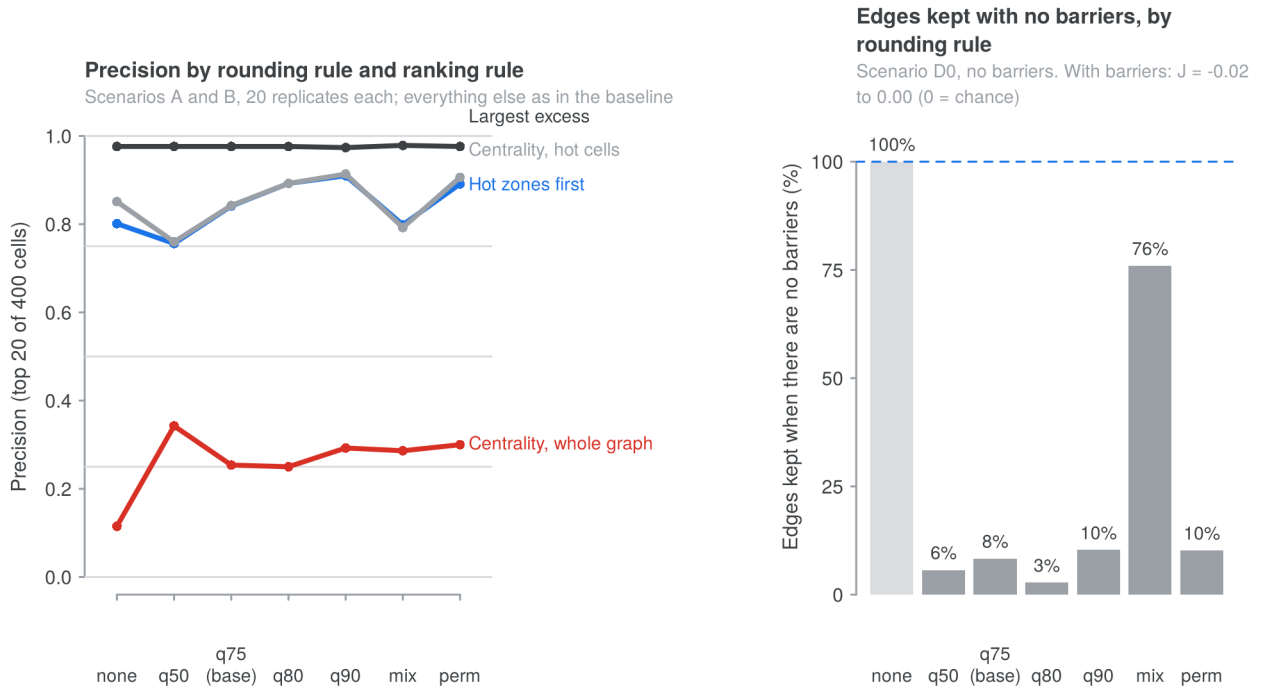


Figure 12: Left: precision of the top 20 of 400 cells by rounding rule and ranking rule (scenarios A and B pooled, 20 replicates each). Right: share of edges kept in scenario D0, where no edge should be removed. Rules: none (full grid), q50–q90 (quantile of the positive b_{ij}; q75 is the baseline), mix (two-component mixture), perm (permutation).

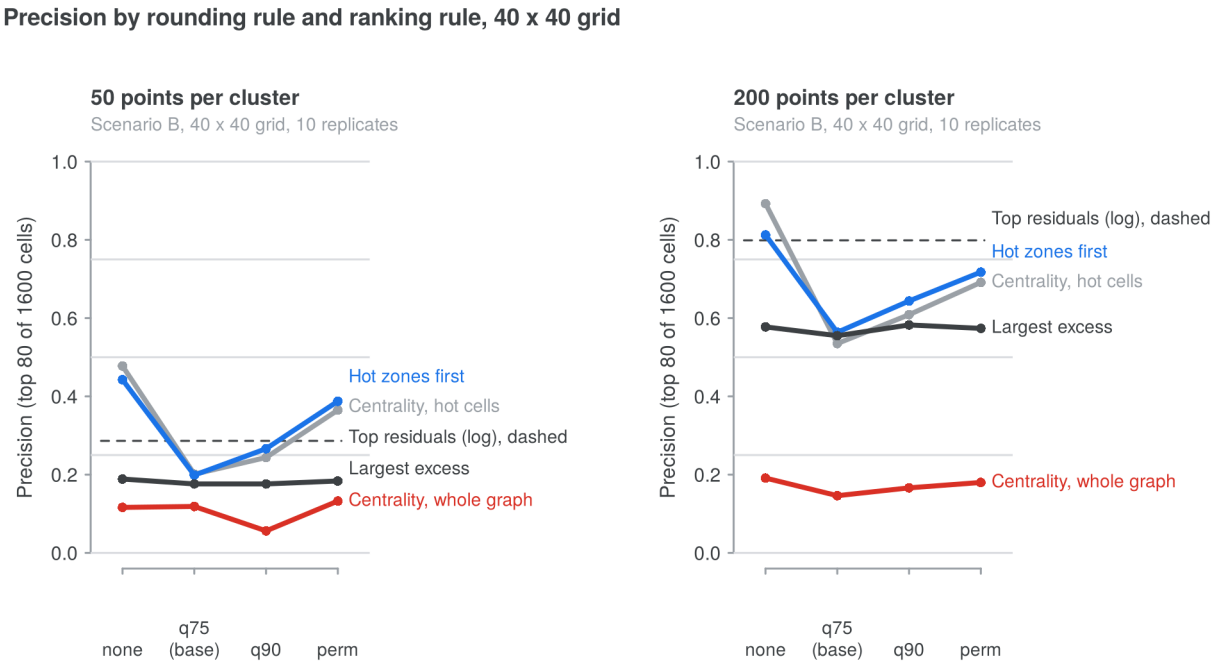


Figure 13: Precision of the top 80 of 1600 cells by rounding rule and ranking rule on a 40×40 grid (scenario B, 10 replicates per panel). Dashed line: top residuals of $\log(Y + 1)$ on Z. Rule labels as in Figure 12.

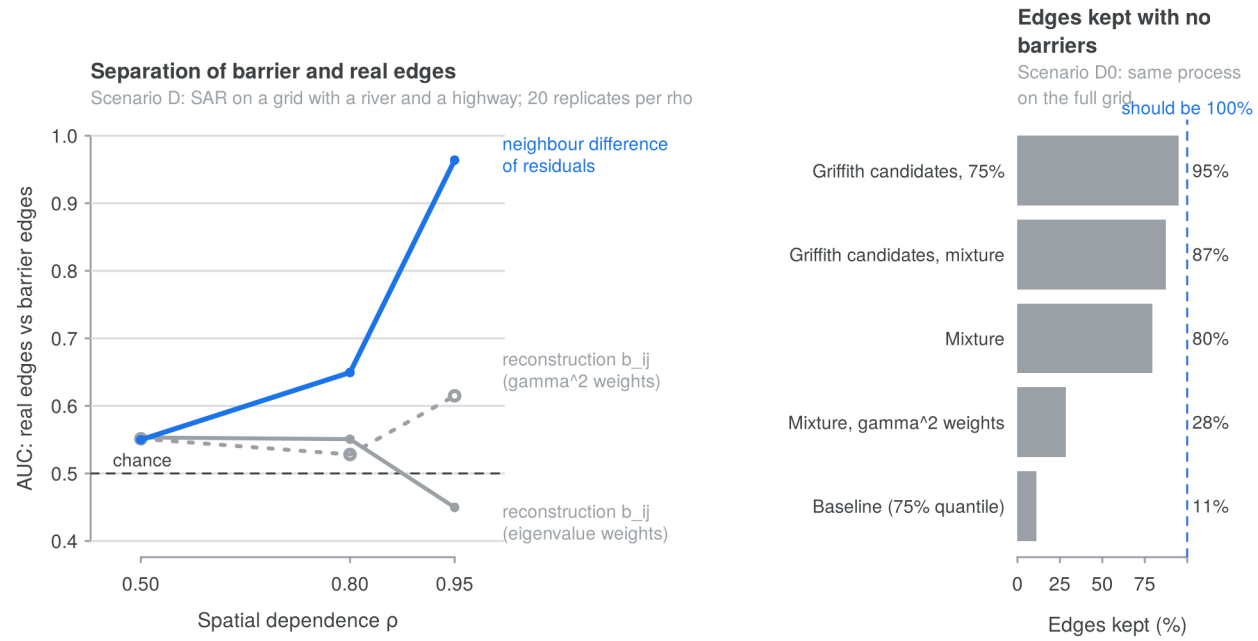


Figure 14: Left: AUC for separating real edges from barrier edges in scenario D (20 replicates per ρ). Right: share of edges kept in scenario D0 by variant of the procedure.